SARSA Policy and Its Explanation in Trading

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Tóm tắt*: Reinforcement Learning (RL) has attracted increasing attention in the field of automated stock trading due to its ability to optimize sequential decision-making and adapt to dynamic environments. Among RL methods, Deep SARSA is an on-policy approach with high stability, making it suitable for noisy financial data. However, like many Deep RL models, Deep SARSA still lacks transparency, which makes its decision-making process difficult to explain and verify in practical financial applications. This study proposes an Explainable Reinforcement Learning (XRL) framework to clarify the decision-making mechanism of a Deep SARSA agent in the stock trading problem. The proposed framework integrates three complementary explanation methods, including Reward Difference Explanation (RDX), Minimal Sufficient Explanation (MSX), and SHapley Additive exPlanations (SHAP). Specifically, RDX and MSX decompose and identify the core reward components that govern action decisions, while SHAP is used to quantify the contribution magnitude and directional impact of input features on the model's Q-values. Experiments are conducted on the Vietnamese stock market dataset (VNStock) using six representative stocks, namely ACB, FPT, GAS, HPG, SSI, and VCB, across three overlapping market periods (2013–2017, 2015–2019, and 2017–2021), reflecting different market conditions ranging from stable growth and market divergence to explosive growth. The results show that the Deep SARSA agent tends to make decisions primarily based on profit expectations and portfolio states during favorable market periods, while its ability to identify and respond to risk remains limited in less stable market contexts. The MSX and SHAP methods provide concise, consistent, and practically valuable explanations, enabling the identification of both rational behaviors and potential weaknesses of the agent. This study contributes to enhancing the transparency and reliability of RL-based trading systems and opens a feasible direction for the application of XRL in the financial domain.

Add (Abstract) at the end of summary

Từ khóa: Explainable Artificial Intelligence (XAI); Reinforcement Learning; Deep SARSA; Automated Trading; VNStock; Reward Decomposition Explanation (RDX); Multi-Step Explanation (MSX); SHAP

I. Introduction

In recent years, the rapid development of data and Artificial Intelligence (AI) techniques has driven significant transformation in the financial and banking sector. AI-based automated trading systems have increasingly demonstrated their effectiveness due to their ability to process large-scale data, identify complex patterns, and optimize decisions at speeds far exceeding human capabilities [1]. In this context, RL has emerged as a promising approach, enabling the modeling of sequential decision-making processes through direct interaction with the market and the optimization of cumulative rewards [2]. In particular, Deep SARSA, an on-policy RL algorithm, has been shown to exhibit high stability and strong capability in learning trading strategies in noisy and uncertain financial environments [3]. However, like most Deep RL methods, Deep SARSA remains inherently a "black-box" model [4]. Although it can make buy, sell, or hold decisions with high accuracy, the model does not provide clear explanations for why a particular action is chosen at each time step. This limitation poses practical challenges, including: (i) investors find it difficult to trust the system when the decision-making mechanism is not transparent; (ii) financial institutions struggle to meet accountability requirements when risks or losses occur; and (iii) the analysis, debugging, and improvement of the model become less effective.

Notably, several recent studies, such as Kumar et al. [4], have employed SHAP to explain the contribution of input features to Q-values in stock trading tasks. This approach is important for enhancing the transparency of Deep RL models at the feature level. However, in trading problems, understanding why an agent prefers one action over another at a given time step requires a deeper perspective that is closely tied to the reward structure, which is the central element of the RL learning mechanism. A promising research direction arises from reward decomposition-based explanation mechanisms, particularly RDX and MSX, proposed by Juozapaitis et al. [5]. In this framework, RDX identifies which reward components support or oppose the selection of one action compared to another. MSX further compresses these reasons into the smallest set that is still sufficient to "certify" the optimality of the chosen action. Although these methods were originally introduced within the DRQ framework, they are inherently model-agnostic and can be applied to any RL agent, provided that the reward function is decomposed into meaningful components. This property enables the direct integration of RDX and MSX with Deep SARSA, allowing agent decisions to be explained at the action level, thereby complementing SHAP explanations at the feature level.

This study makes the following key contributions:

- Proposes a multi-level explainability framework that integrates RDX, MSX, and SHAP for the Deep SARSA model.
- Develops a comprehensive visualization toolkit to support the analysis of agent behavior at both the reward level (RDX, MSX) and the feature level

(SHAP), helping analysts better understand decision drivers and the model's learning mechanism.
- Constructs a systematic experimental pipeline to evaluate and compare the model's explainability across different market scenarios (bullish, bearish, sideways).
- Provides a pioneering contribution in Vietnam on XAI for Deep RL in finance.

The remainder of this paper is organized as follows. Section II provides a comprehensive review of related work on Explainable AI in finance and the current state of Explainable Reinforcement Learning. Section III details the proposed methodology, describing the integration of the Deep SARSA agent with the RDX, MSX, and SHAP explanation modules. Section IV presents the experimental setup, including the VNStock dataset and training configuration, followed by a detailed analysis of the empirical results and the evaluation of the explanations. Finally, Section V summarizes the key findings and provides direction for future research.

II. Literature Review

A. Explainable AI (XAI) in Finance

Explainable Artificial Intelligence (XAI) is becoming an essential component of modern financial systems, where requirements for transparency, auditability, and accountability are always prioritized. International regulatory bodies such as the Bank of England and the Financial Conduct Authority (FCA) have emphasized that AI models in finance must provide clear, verifiable explanations and comply with risk governance standards in highly regulated environments [6]. This highlights that understanding a model's decision-making mechanism is not only a technical concern but also directly related to ethical issues, market trust, and regulatory compliance [7]. Among contemporary XAI methods, SHAP has emerged as a standard technique due to its strong theoretical foundation in Shapley values from game theory, which ensures important properties such as consistency and additivity [8]. SHAP has been widely applied in various financial tasks, including credit scoring, risk analysis, and market forecasting, thanks to its ability to transparently and mathematically quantify the contribution of each feature [9]. Compared to traditional techniques, SHAP provides more stable and coherent interpretations, thereby increasing trust in highly complex AI models.

However, most existing XAI techniques primarily focus on supervised learning models and have not yet received proportional attention from an explainability perspective [10]. Traditional XAI methods often fail to effectively handle the sequential nature, strong temporal dependence, and long-horizon optimization objectives that are characteristic of RL [11]. In addition, the exploration–exploitation mechanism and delayed rewards further make it particularly challenging to trace the reasons behind an agent's actions. Therefore, in the financial context, where each trading decision has a direct impact on risk and profit, there is an urgent need to develop XAI methods specifically designed for RL, capable of explaining not only input signals but also action strategies and reward structures over time [12]. This represents a critical research gap that recent XRL studies aim to address, yet it remains underexplored in the domain of automated trading.

B. Explainable Reinforcement Learning

XRL is an important research area aimed at enhancing the interpretability of RL agents, which are characterized by learning processes that depend on states, actions, and rewards over time and are more difficult to interpret than supervised learning models. XRL seeks to explain three core components of RL: (i) the action policy; (ii) the Q-value function; and (iii) the reward structure factors that determine the agent's long-term strategy [13]. This need is particularly critical in high-risk domains such as finance, autonomous driving, and robotics, where transparency and verifiability are increasingly emphasized. The main approaches in XRL can be categorized into the following three prominent groups:

- Reward Difference Explanation (RDX): This method decomposes the total reward into meaningful reward components and compares the contribution of each component to the value difference between two actions [5]. This approach not only helps identify which components—such as profit, risk, or transaction costs—drive the agent to prefer one action over another, but also provides a structured explanation grounded in the fundamental nature of RL.
- Minimal Sufficient Explanation (MSX): This method is based on selecting the minimal set of reasons that is still sufficient to explain why action a_1 is preferred over a_2. MSX identifies the smallest set of positive reward components (MSX^+) and the key negative reasons (MSX^-) that make the positive set necessary [5].
- Feature Attribution for Q-values: Techniques such as SHAP-RL apply Shapley values to quantify the influence of each input feature on the Q-value of each action [8]. Unlike RDX/MSX, which provide explanations in the reward space, SHAP offers explanations in the input feature space, enabling the identification of important market signals that the agent uses to make trading decisions.

Although XRL has made significant progress, the field still faces many challenges, such as the dynamic and non-stationary nature of RL environments, delayed rewards that make causal tracing difficult, the combination of exploration and exploitation causing agent behavior not to always reflect optimal values, and the increasing complexity of modern Deep RL architectures [14]. These challenges highlight the need for XRL methods specifically designed for financial trading problems, where transparency, reliability, and accurate explainability are of paramount importance.

C. Applications of XAI in Financial Reinforcement Learning

The application of XAI to RL models in financial trading has become a promising research topic, especially as RL models are increasingly used to develop automated trading strategies. Many recent studies have experimented with integrating XAI into agents such as DQN, PPO, and A2C to visualize policies, analyze the importance of market features, or evaluate the evolution of Q-values over time [15]. These studies demonstrate that XAI can significantly enhance the transparency of trading systems, support model auditing, reduce operational risk, and help investors better understand the agent's decision-making mechanisms. However, this field still contains several important research gaps. Most existing XRL studies in finance focus on developed markets such as the United States and Europe [16], where data are relatively stable and liquidity is high. In contrast, emerging markets such as Vietnam, which are characterized by high volatility, lower liquidity, and strong influence from investor sentiment, have been largely underexplored from an XRL perspective [17]. This limitation restricts the ability to assess the practical applicability of explainability methods in more complex market environments.

More importantly, no prior study has combined RDX and MSX to explain the behavior of a Deep SARSA agent in the context of financial trading. Existing works either focus on feature-level explanations through SHAP or investigate popular agents such as DQN and PPO. In addition, no study has proposed a multi-level XAI framework that integrates reward-based explanations (RDX, MSX) and feature-based explanations (SHAP) for RL in finance. These gaps highlight the necessity of a study capable of providing comprehensive explanations that are well aligned with the characteristics of RL and the context of emerging markets. Therefore, this research aims to fill this gap by integrating advanced XAI techniques including RDX, MSX, and SHAP into the Deep SARSA model and evaluating their effectiveness on the Vietnamese stock market dataset (VNStock). This represents a novel approach that has not been previously explored and has the potential to make a significant contribution to the development of transparent and trustworthy automated trading systems for the Vietnamese financial market.

III. METHODOLOGY

A. Proposed XAI–RL Framework

This study proposes a two-layer XAI–RL framework to simultaneously optimize the trading policy and provide transparent explanations for the agent's behavior. The framework consists of: (i) a Deep SARSA agent layer, responsible for learning an optimal trading policy through interaction with the VNStock market and updating the Q-function using a deep neural network; and (ii) an XAI module layer, which integrates RDX, MSX, and SHAP to analyze and interpret the factors governing the agent's decisions.

Figure 1 details the workflow for Explaining Deep SARSA in our system. The workflow commences with the training of a "black box" Deep SARSA agent on historical stock data during favorable market phases.

B. Reward Decomposition Explanation

To explain why the agent prefers action a_1 over action a_2 in state s ($Q(s, a_1) > Q(s, a_2)$), this study employs RDX. This method analyzes the difference between the two actions based on decomposed reward components, thereby providing a transparent explanation of which factors lead to the agent's preference. First, the total reward at time t is decomposed into four financially meaningful components:

$$r_t = r_t^{\text{profit}} + r_t^{\text{risk}} + r_t^{\text{position}} + r_t^{\text{stability}}, \quad (1)$$

This decomposition allows the model to learn action values in the form of a decomposed Q-vector, rather than only a scalar aggregated Q-value. Specifically, for each action a, the agent maintains a vector:

$$\mathbf{Q}(s, a) = [Q_c(s, a)]_{c \in C}, \quad (2)$$

In this vector, each component $Q_c(s, a)$ represents the contribution of reward type c to the long-term value of the action. For example, Q_{profit} reflects expected profit, Q_{risk} represents anticipated risk, Q_{position} indicates trend alignment, and $Q_{\text{stability}}$ measures price volatility. RDX is defined as the difference between the two decomposed Q-vectors corresponding to the two actions:

$$\Delta(s, a_1, a_2) = \mathbf{Q}(s, a_1) - \mathbf{Q}(s, a_2), \quad (3)$$

or, more specifically, for each individual component:

$$\Delta_c(s, a_1, a_2) = Q_c(s, a_1) - Q_c(s, a_2). \quad (4)$$

The meaning of each value Δ_c is as follows:

- If $\Delta_c > 0$, reward component c supports choosing action a_1.
- If $\Delta_c < 0$, component c opposes action a_1 and favors action a_2.

C. Minimal Sufficient Explanation

Although RDX provides a complete description of all reward components that influence the difference between two actions, the number of reasons can become large and difficult to interpret. Therefore, this study employs MSX to identify the minimal set of the most important reasons that is still sufficient to preserve the action preference ordering produced by the agent. MSX thus serves as a concise yet accurate certificate for explaining the agent's behavior. At state s, the MSX explanation for preferring action a_1 over a_2 is represented as a pair:

$$(\text{MSX}^+, \text{MSX}^-), \quad (5)$$

where:

- MSX^+ is the set of core positive reasons explaining why the agent prefers a_1;

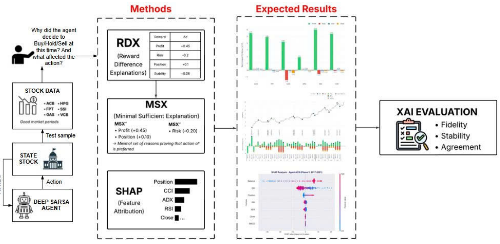

Figure 1: Overall XAI-–RL Workflow for Explaining Deep SARSA Trading Agent.

- MSX^- is the set of core negative reasons that confirm that every element in MSX^+ is necessary.

Based on the decomposed reward difference values $\Delta_c(s, a_1, a_2)$, the **disadvantage** of action a_1 relative to a_2 is defined as follows:

$$d = \sum_c I[\Delta_c(s, a_1, a_2) < 0] \cdot |\Delta_c(s, a_1, a_2)|, \quad (6)$$

where:

- $\Delta_c(s, a_1, a_2) = Q_c(s, a_1) - Q_c(s, a_2)$;
- $I[\cdot]$ denotes the indicator function.

The value d reflects the total magnitude of negative influences, that is, the reward components that support choosing a_2 instead of a_1. MSX^+ is defined as the smallest subset of positive reasons ($\Delta_c > 0$) such that their total influence exceeds the disadvantage d:

$$\text{MSX}^+ = \arg\min_{M \subseteq C} |M| \quad \text{s.t.} \quad \sum_{c \in M} \Delta_c(s, a_1, a_2) > d. \quad (7)$$

This definition ensures that MSX^+ includes only the reasons that are truly necessary to explain why a_1 is preferred. If multiple sets satisfy this condition, the set with the larger total influence is preferred; if a tie remains, lexicographic ordering is applied. MSX^- answers the question: "Why are all reasons in MSX^+ necessary?" To do this, the just-insufficient advantage is computed as follows:

$$v = \sum_{c \in \text{MSX}^+} \Delta_c(s, a_1, a_2) - \min_{c \in \text{MSX}^+} \Delta_c(s, a_1, a_2). \quad (8)$$

The value v represents the total advantage of a_1 after removing the weakest positive reason in MSX^+. Based on this value, MSX^- is defined as the smallest set of negative reasons ($\Delta_c < 0$) whose total disadvantage exceeds v:

$$\text{MSX}^- = \arg\min_{M \subseteq C} |M| \quad \text{s.t.} \quad \sum_{c \in M} -\Delta_c(s, a_1, a_2) > v. \quad (9)$$

This set demonstrates that if any positive reason in MSX^+ is removed, the remaining advantage of a_1 would no longer be sufficient to overcome the core negative reasons, indicating that every component in MSX^+ is truly necessary.

D. Feature Attribution with SHAP

To explain the influence of each technical feature on trading decisions, this study implements the SHAP method based on the foundational work of Lundberg and Lee [8]. This approach provides contribution values of each feature to the Q-value of the selected action.

General SHAP Formula: The SHAP value for the i-th feature is computed according to the following formula:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [Q(s, S \cup \{i\}) - Q(s, S)], \quad (10)$$

where:

- F: denotes the set of all features;
- S: denotes a subset of features;
- $Q(s, S)$: is the Q-value computed using only the feature subset S;
- $|S|$: is the number of features in set S.

For the optimal action a^*, the SHAP value is determined as follows:

$$\phi_i(a^*) = \text{contribution of feature } x_i \text{ to } Q(s, a^*) \quad (11)$$

This method allows answering important questions regarding the influence of technical indicators:

- RSI: "How does the current RSI level affect the Buy decision?"
- MACD: "Does MACD increase or decrease the Q-value of the action?"
- ADX: "Does ADX have a strong or weak influence on Sell behavior?"

IV. Evaluation

A. Evaluation Schema

The evaluation scheme is designed to simultaneously verify the trading performance of the Deep SARSA agent and the quality of XAI explanations (RDX, MSX, SHAP). Therefore, the evaluation process relies not only on traditional trading performance metrics but also considers the transparency, consistency, and interpretability of the agent's behavior. The scheme consists of three main components: the dataset, the training configuration, and the evaluation metrics system.

Dataset: The study uses historical data from six large-cap and highly liquid stocks in the Vietnamese stock market (ACB, HPG, FPT, SSI, GAS, and VCB), which represent key industry sectors and are frequently included in the VN30 index basket [18], thereby providing a relatively comprehensive reflection of VN-Index dynamics [19]. The selection of blue-chip stocks helps ensure data quality and enhance the reliability of experimental results. The data are divided into three overlapping experimental blocks (Table I), each consisting of a training period and a subsequent evaluation period, in order to assess the generalization ability of the Deep SARSA agent under different market conditions and reduce the impact of abnormal shocks. In addition, the selected stocks exhibit diverse volatility characteristics across sectors, supporting the evaluation of the robustness of both the trading policy and the XAI explanations. The training configuration of the Deep SARSA agent in this study is summarized in Table II.

Table I: Experimental blocks used in the study

Block	**Training period**	**Test period**
Block 1	2013–2016	2017
Block 2	2015–2018	2019
Block 3	2017–2020	2021

Experimental Design: The experimental design in this study is constructed to simultaneously analyze trading behavior and the explanations generated by the Deep SARSA agent across different market growth phases. For each experimental block, the Deep SARSA agent is first trained on the corresponding training period. The trained model is then applied to the subsequent evaluation period

Table II: Experimental configuration of the Deep SARSA trading agent.

Component	Configuration
Algorithm	Deep SARSA (on-policy)
Actions	Buy, Sell, Hold
Q-network architecture	Two fully connected hidden layers (ReLU)
Exploration strategy	ε-greedy
Reward function	R_{profit}, R_{risk}, R_{position}, $R_{\text{stability}}$
Number of episodes	35 per agent

to execute trades and record Buy/Sell/Hold decisions. During the evaluation phase, at each decision time step, the explanation methods RDX, MSX, and SHAP are activated to analyze the selected action. Specifically, RDX and MSX are used to clarify the reward-based reasons that govern action preference, while SHAP is applied to identify the contribution of input features to the Q-value of the corresponding action. This procedure is repeated for all stocks and experimental blocks, enabling a comparison of the Deep SARSA agent's behavior and explanation structures under different market contexts.

B. Reward Difference Explanation

In this section, the study presents RDX explanation results at a specific time t during the testing phase to analyze the decision-making mechanism of the Deep SARSA agent after training. Selecting a time point from the test set allows the evaluation of the model's generalization ability in a context that has not appeared during training. At each time step, the Q-values $Q(s_t, a)$ are predicted over the entire action space and decomposed into four reward components: Profit, Risk, Position, and Stability to generate RDX explanations.

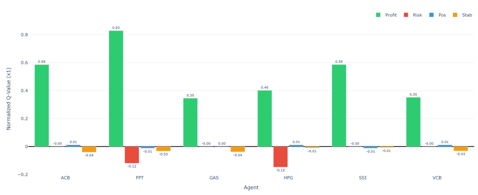

Figure 2: RDX results at a specific time t during the testing phase (2017) of the Deep SARSA agent on six stocks: ACB, FPT, GAS, HPG, SSI, and VCB.

Figure 2 shows that the Profit component clearly dominates the agent's decision structure, particularly for SSI, ACB, and VCB, indicating that high profit expectations are the primary drivers of the selected actions. In contrast, Risk and Stability are predominantly negative, representing unfavorable factors considered by the agent during the decision-making process; notably, GAS exhibits a large negative Risk value, reflecting a cautious stance toward volatility specific to the energy sector. The Position and Stability components generally contribute only marginally, serving as secondary adjustment factors rather than decisive elements. Overall, the RDX results indicate that Deep SARSA makes decisions based on a reasonable trade-off between expected profit and unfavorable factors, while the explanations consistently reflect the volatility characteristics of each stock within the same market context, thereby confirming the transparent explanatory capability of RDX during the testing phase.

C. Minimal Sufficient Explanation

Explanations (MSX) of the Deep SARSA agent during the testing phase. The MSX plots are constructed at predefined critical points, including periods of sharp declines, trend reversals, and abrupt changes in the agent's actions. Focusing on these key moments allows a clear analysis of the core reason structure that governs the agent's decisions in situations with high uncertainty.

Period 2013—2017: The 2017 testing period reflects a market context with relatively stable growth accompanied by local correction phases. In this setting, the MSX results indicate that the decision-making behavior of the Deep SARSA agent is generally consistent with investment logic in an uptrend market (Figure 3). At critical points corresponding to strong correction phases, the agent tends to prioritize high-intensity buy actions, primarily driven by the dominant Profit component, while Risk and Stability play only secondary roles. This suggests that the agent has learned a "buy-the-dip" strategy to maximize returns during an upward trend.

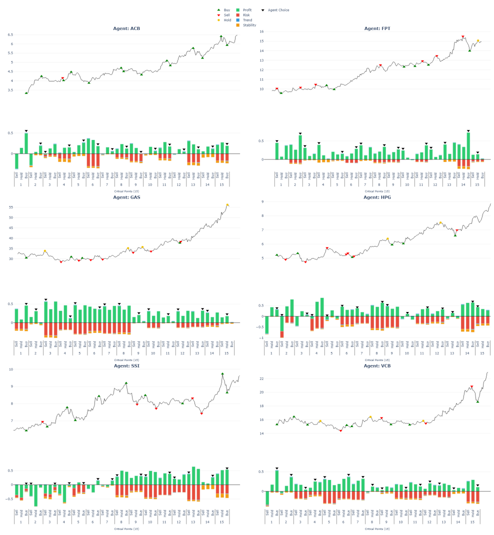

Figure 3: MSX explanation results at critical points during the 2017 testing period for the six stocks ACB, FPT, GAS, HPG, SSI, and VCB.

In contrast, at trend reversal points or the formation of local peaks, the agent tends to switch to sell actions when the expected profit from continued holding declines, reflecting short-term profit-taking behavior. However, MSX also reveals several suboptimal decisions, in which the agent continues to prioritize actions primarily based on profit while insufficiently accounting for risk factors. Overall, the 2017 MSX results indicate that the agent performs effectively in a growth-oriented market but remains limited in its ability to balance profit and risk.

Period 2015—2019: The 2019 testing period reflects a market context with strong divergence across stocks. The MSX results (Figure 4) show that the Deep SARSA agent performs effectively on stocks that maintain upward trends, such as FPT and VCB, where buy actions at critical points are primarily driven by the dominant Profit component, yielding positive trading outcomes despite the presence of Risk factors. This indicates that the agent's is capable of trend-following when market conditions are favorable.

Figure 4: MSX explanation results at critical points during the 2019 testing period for the six stocks ACB, FPT, GAS, HPG, SSI, and VCB.

In contrast, for stocks with sideways or declining trends such as HPG and SSI, the agent continues to apply the "buy-the-dip" strategy. MSX analysis shows that these decisions are still primarily based on profit expectations, while the Risk component is undervalued or ignored, leading to many ineffective trading actions and losses. In addition, for GAS, MSX highlights several unreasonable decisions in which the agent prioritizes actions based on unstable profit forecasts and secondary components, reflecting a misalignment between the predicted Q-values and actual market movements. Overall, the MSX results for 2019 indicate that the Deep SARSA agent is unable to generalize effectively to divergent or declining market contexts. The MSX explanations show that the agent tends to over-rely on the Profit component, while its ability to recognize and adapt to risk remains limited.

Period 2017—2021: The 2021 testing period reflects a context of strong and prolonged market growth. The MSX results (Figure 5) show that the Deep SARSA agent adapts well to this favorable environment, particularly at critical points corresponding to short-term corrections, where the agent often prioritizes high-intensity buy actions. For VCB, HPG, and ACB, MSX indicates that the Profit component clearly dominates over Risk, enabling the agent to effectively exploit recovery phases and demonstrating a consistently effective "buy-the-dip" strategy.

For SSI, the agent primarily maintains a buy-and-hold strategy throughout the growth period, with buy decisions driven by positive Profit, despite higher volatility. Meanwhile, GAS exhibits more cautious behavior, with alternating buy and sell actions at critical points, reflecting the uneven nature of its upward trend. In contrast, for FPT, MSX reveals several inefficient decisions, where the agent sells too early while the upward trend remains strong, resulting in missed subsequent gains. The MSX results for 2021 indicate that the Deep SARSA agent performs effectively on VCB, HPG, and ACB, with decisions largely driven by the Profit component. However, the MSX explanations also reveal inconsistencies for FPT, GAS, and SSI, particularly in identifying optimal exit timing and managing risk under conditions of strong market growth and high volatility.

Figure 5: MSX explanation results at critical points during the 2021 testing period for the six stocks ACB, FPT, GAS, HPG, SSI, and VCB.

D. SHapley Additive exPlanations

To complement the input feature level perspective and clarify which factors govern the Q-values during the decision-making process, this study further employs the SHAP method. Analyzing SHAP plots across different testing periods helps reveal changes in the roles of technical indicators and trading states in the decision-making mechanism of the Deep SARSA agent as market conditions evolve.

Period 2013—2017: The 2017 testing period (Figure 6) shows that the decision-making mechanism of the Deep SARSA agent is strongly influenced by several key input features, particularly portfolio states (Balance, Position) and momentum based technical indicators such as CCI and RSI. This reflects a good alignment between the agent's decision structure and the relatively stable upward market trend during this period. Specifically, for ACB, GAS, and HPG, the SHAP values of CCI and

Position are widely distributed in both directions, indicating that overbought/oversold signals and holding states play an important role in driving buy or sell decisions. Meanwhile, Balance often exhibits large positive SHAP values, especially for FPT, SSI, and VCB, suggesting that available capital is a key factor influencing the intensity of the agent's actions in an uptrend market.

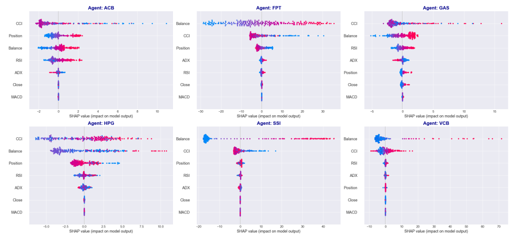

Figure 6: Bee swarm plots showing the results of the agents during the 2013–2017 period.

In contrast, long-term trend indicators such as ADX, MACD, and the closing price (Close) have SHAP values concentrated near zero for most stocks, indicating a relatively limited role in the decision-making process. This implies that during 2017, the Deep SARSA agent tended to rely more on short-term signals and internal portfolio states rather than on slow-moving trend indicators. Overall, the SHAP results for 2017 show that the agent's behavior is primarily driven by factors reflecting market momentum and position expansion capacity, which is consistent with a bullish market environment. At the same time, SHAP analysis also highlights an imbalance in feature contributions, suggesting that the model still has limitations in effectively leveraging risk-related information and long-term trend signals.

Period 2015—2019: Entering the 2019 testing period, the SHAP results (Figure 7) show that the decision-making dynamics of the Deep SARSA agent become clearly diversified across stocks, directly reflecting the heterogeneous market conditions of that year. Compared to 2017, the contribution of input features is no longer stable but varies substantially across individual stocks, indicating an increasing dependence of the agent's behavior on the local conditions of each price series. For FPT and VCB, the portfolio state (Balance) remains the most dominant feature, exhibiting large positive SHAP values when available capital is high, suggesting that the agent tends to increase action intensity during continued price increases. At the same time, momentum indicators such as CCI and ADX also make significant contributions, reflecting the agent's effective exploitation of trend signals in stocks that maintain upward momentum during 2019.

In contrast, for HPG and SSI, the distribution of SHAP values reveals a clear misalignment between expectations and actual market conditions. Specifically, Balance continues to exhibit very large positive SHAP values, while risk and trend indicators (such as ADX and RSI) have limited impact and are often concentrated near zero. This indicates that the agent continues to make decisions based on its capacity to expand positions, even though technical signals no longer support an upward trend,

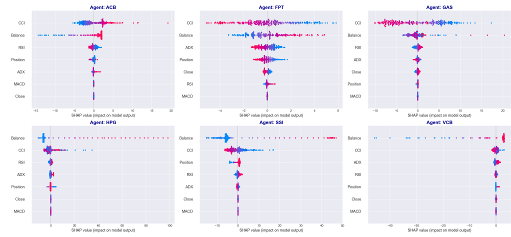

Figure 7: Bee swarm plots showing the results of the agents during the 2015–2019 period.

leading to poor performance in sideways or declining market environments. For ACB and GAS, the SHAP plots show a more moderate level of influence, with CCI and Balance remaining the primary drivers but with smaller magnitudes. This reflects more cautious decision-making behavior by the agent in the presence of strong fluctuations and a lack of clear price trends.

Overall, the SHAP results for 2019 indicate that the Deep SARSA agent relies heavily on capital state and short-term momentum signals, while its ability to adjust feature weighting in response to changes in market structure remains limited. The SHAP analysis clarifies the causes of performance differences across stocks and highlights the agent's difficulty in recognizing and adapting to divergent and trendless market conditions.

Period 2017—2021: The SHAP results for the 2021 testing period (Figure 8) show that the Deep SARSA agent operates in a strong growth market context, with a markedly increased role of the portfolio state (Balance) across most stocks. For VCB, HPG, and GAS, Balance is the dominant feature with large positive SHAP values, reflecting the agent's strong utilization of available capital to increase action intensity during upward trend phases. At the same time, momentum indicators such as CCI and ADX make significant contributions, indicating the agent's effective exploitation of trend signals during the explosive uptrend period.

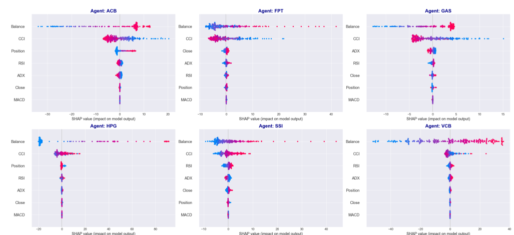

Figure 8: Bee swarm plots showing the results of the agents during the 2017 – 2021 period.

For FPT, although the upward trend is still maintained, the SHAP distribution shows that the agent is jointly influenced by Balance and CCI, but with lower intensity, reflecting a more cautious decision-making tendency after strong price increases. Meanwhile, ACB and SSI exhibit a moderate level of influence, where Balance remains the dominant factor while indicators such as CCI and RSI begin to contribute more noticeably, indicating that the agent flexibly combines portfolio state and technical signals. Overall, the SHAP results for 2021 indicate that the key strength of the Deep SARSA agent lies in its ability to effectively optimize decisions based on capital state and market trends under strong growth conditions. However, its weakness is reflected in the dominance of Balance, suggesting a decision-making strategy that amplifies profit and remains limited in leveraging risk and stability indicators, making performance highly dependent on favorable market conditions.

V. Conclusion

This study proposes an Explainable Reinforcement Learning (XRL) framework to clarify the decision-making mechanism of a Deep SARSA agent in stock trading. The integration of RDX, MSX, and SHAP enables simultaneous analysis of trading performance and decision-making motivations under different market contexts. Experiments on six Vietnamese stocks show that the agent's behavior is mainly driven by profit expectations and portfolio state during market growth periods, while in divergent market phases, MSX and SHAP explanations reveal imbalances in the decision structure and limitations in risk recognition.

The main contribution of this study is demonstrating that post-hoc explanation methods such as MSX and SHAP can provide concise, consistent, and practically valuable explanations for deep reinforcement learning models. However, due to their post-hoc nature, the agent has not yet learned multi-objective risk management strategies during training. Future work will focus on integrating explainability directly into the model, extending evaluation to crisis market periods, and incorporating macroeconomic indicators.

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